

Spencer D.H. Pruno

+81 070 546 02644

LinkedIn: <https://www.linkedin.com/in/spencerfx>

Email: spencer.pruno@gmail.com

PROFESSIONAL EXPERIENCE

Data Scientist (Financial Risk Models)

[Yorii – Tokyo, Japan; *May 2026 –*]

▫ **Financial Risk Models - Credit**

- Co-leading machine learning algorithm modeling for credit score & cash flow analysis on SMEs / Startups
- Partner with Financial Risk Analysts to design new risk analysis tools and implement new data pipelines

KDB+ Analytics Developer (Senior Technology Associate)

[UBS via Synechron – Hong Kong, SAR; *June 2025 – December 2025*]

▫ **Equity Analytics:**

- Provided strategic analysis and implementation for equity analytics on algorithmic trading strategies
- Re-wrote and functionalized code for all core processes, so that new modules could be written efficiently and with less code. Reduced maintenance cost while improving readability and modularity

▫ **Architecture, Stability and Performance improvements**

- Designed and wrote all the kdb+ system and performance monitoring software
- Added coverage for memory, logging and latency information. Used the new tools to identify all performance issues within the kdb plant. Led process of performance improvement discovery.

▫ **Reference Data SME**

- Designed the order data enrichment module and reference data service within the platform for alerts
- API / Gateway development so that front-end developers could use this data

Senior Software Engineer

[MUFG UFJ via Karabiner Software – Tokyo, Japan; *May 2023 – April 2025*]

▫ **FX Analytics Development**

- Re-wrote and designed various PnL, risk Analytics Algorithms for performance increase (~300%)
- Enhancements to custom CEPs for deal analytics (mark-outs, impact, TCA) as SME for analytics data
- Wrote new ingestion framework for external deal/algorithmic order flow analytics (RT file-based FH)
- Designed and architected cloud-based JupyterNotebook/Python integration for quants and sales traders

▫ **Leadership and Core KDB Development**

- Developed automated intraday recovery and write-down solutions (in KDB and Python)
- Real-time and Historical system telemetry via customized functions and dashboards (Grafana)
- Developed an Automated Testing Framework utilizing KDB/Python with visualizations via Jenkins
- Led design and development of improved security administration/API for whole KDB platform
- Added strategic enhancements to gateways via design changes for performance and access improvements

Software Engineer (FBOT - Archers Squad)

[Fidelity via Beacon Hill – California, USA; *Aug. 2021 – May 2023*]

▫ **Credit Analytics**

- Developed analytics for YTM, YTE and Mark-to-Market for various credit assets utilizing KDB
- Re-wrote and supported regulatory-related credit analytics used by Reporting teams
- Designed and developed integration between KDB DB and SQL using Python
- Developed improvements for Java-based feedhandler and REST APIs around KDB data

▫ **Leadership and Core KDB Development**

- Senior Engineer that co-led KDB development within Fixed Income brokerage tech
- Provided Design and Strategy input to improve KDB services for Greenfield Credit trading platform
- Demonstrated operational excellence by improving deployment velocity of KDB services by over 70% YTD and reduced production incidents by 90% YoY
- Expanded system metric intelligence by expanding monitoring across all core KDB platforms

KDB+ Engineer (RADARS Support Lead)

[Standard Chartered – New York, USA; Aug. 2019 – July 2021]

- **Kdb+ Architecture:** Lead kdb+ engineer, managing a team of 7 supporting all major FX, rates, credit products' databases and down-stream consumers. The team is responsible for supporting core market data ETL, as well as specialized applications in execution analytics, regulatory data commitments, algorithmic trading services for clients, PnL analysis and risk management.
- **Quant Front-Office Development:** Collaborated with quants and e-traders in the development of analytics tools
 - **FX Spread Calibration:** Collaborating with senior traders by architecting new spread calibration workflow utilizing python and kdb+ based market data analysis
 - **Analytics Reporting Framework:** Helped lead a PoC to perform / visualize PnL analytics to send to head of trading
 - **Automated Post-Deployment Checks:** Collaborated with support team-members and developers to automate release checks, thereby improving stability and efficiency
- **Git Repo Management:** Created and led kdb_support git repository and led the SDLCC
- **Hardware Management:** Responsible for managing KDB estate encompassing over 50 servers and 700tb of data

KDB+ Engineer

[Virtu (ITG) via First Derivatives – New York, USA; September. 2018 – Jul. 2019]

- **Support Tools:** Built highly functional production support tools spanning all processes of a broker dealer equity trading system. Launched machine learning (SVM) POC to detect anomalies in log files and OMS processes.
- **ITRS / GENEOS:** Worked to implement firm-wide monitoring for all kdb+ processes. Oversaw data quality and job monitoring for reference data, market data and trade data. Adhoc data research for business users.

KDB+ Engineer

[Bank of America via First Derivatives – New York, USA; Jan. 2018 – Sep. 2018]

- **Real-time analysis Tools:** Implemented and wrote Kx Surveillance platform. Provided requirement gathering, design and development for a variety of market micro-structure, order book analytics.

Algorithmic Trading Project Manager

[Forex Capital Markets Ltd. – New York, USA; Feb. 2017 – Dec. 2017]

Level 2 Technology Support

[Forex Capital Markets Ltd. – San Francisco, California; May 2016 – Feb. 2017]

TECHNICAL SKILLS

Programming Languages:

- KDB+/q, Python, Unix (bash), Java, Matlab, SQL

Data Tools:

- DataDog, Datawatch, Grafana, JupyterNotebook (with kdb+), Kx Dashboards, QuantConnect

SLDC / Infra Tools:

- Ansible, AWS, BitBucket, Docker, GitLab, Jenkins, uDeploy, Kubernetes, Selenium

Financial Engineering:

- Market Data Analysis
- Asset Pricing and Risk
- Reporting Regulations
- Market Manipulation Detection
- Natural Language Processing (NLP)
- Liquidity Analysis

EDUCATION

College: BS – Economics – Southern Oregon University (December 2014)

CERTIFICATIONS

- Deep Learning.ai: Neural Networks and Deep Learning, Convolutional Neural Networks, Sequence Models, etc
- Imperial College London: Mathematics for Machine Learning: Linear Algebra, Multivariate Calculus, PCA
- National Futures Association: Series 3 and 34 (Expired)
- Stanford University: Machine Learning, University of Michigan: Introduction to Data Science in Python

VOLUNTEER / CHARITY

- Organization: Two by Two Tutoring (New York City) January 2017 – December 2019
- Organization: The Bowery Mission (New York City) December 2018 – August 2019
- Organization: Ronald McDonald House (New York City) October 2018 – August 2019